Global CO_2 Emissions Who Emits, How Much, and What's Changing

Fossil Fuels & Industry, 1750–2024

CO_2

>35 Bt/yr

Data Source:

Global Carbon Budget 2025 via Our World in Data

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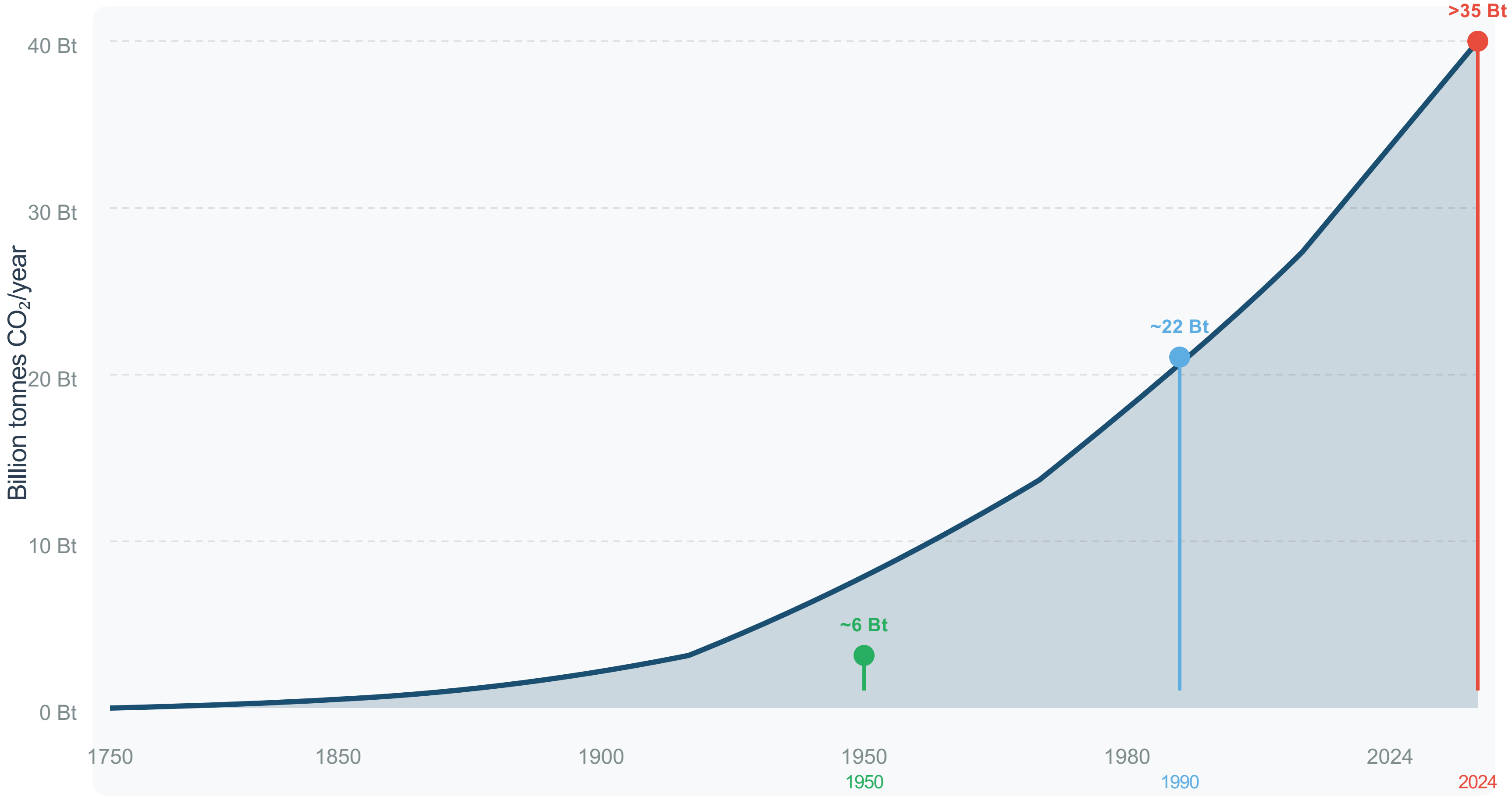
For: Policy Analysts | Climate Negotiators | Sustainability Researchers

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Fossil fuels & industry emissions trajectory, 1750–2024



Key Insights

Pre-Industrial Era

Emissions were negligible
before the Industrial Revolution

Post-1950 Acceleration

6x increase in just 74 years
driven by fossil fuel combustion

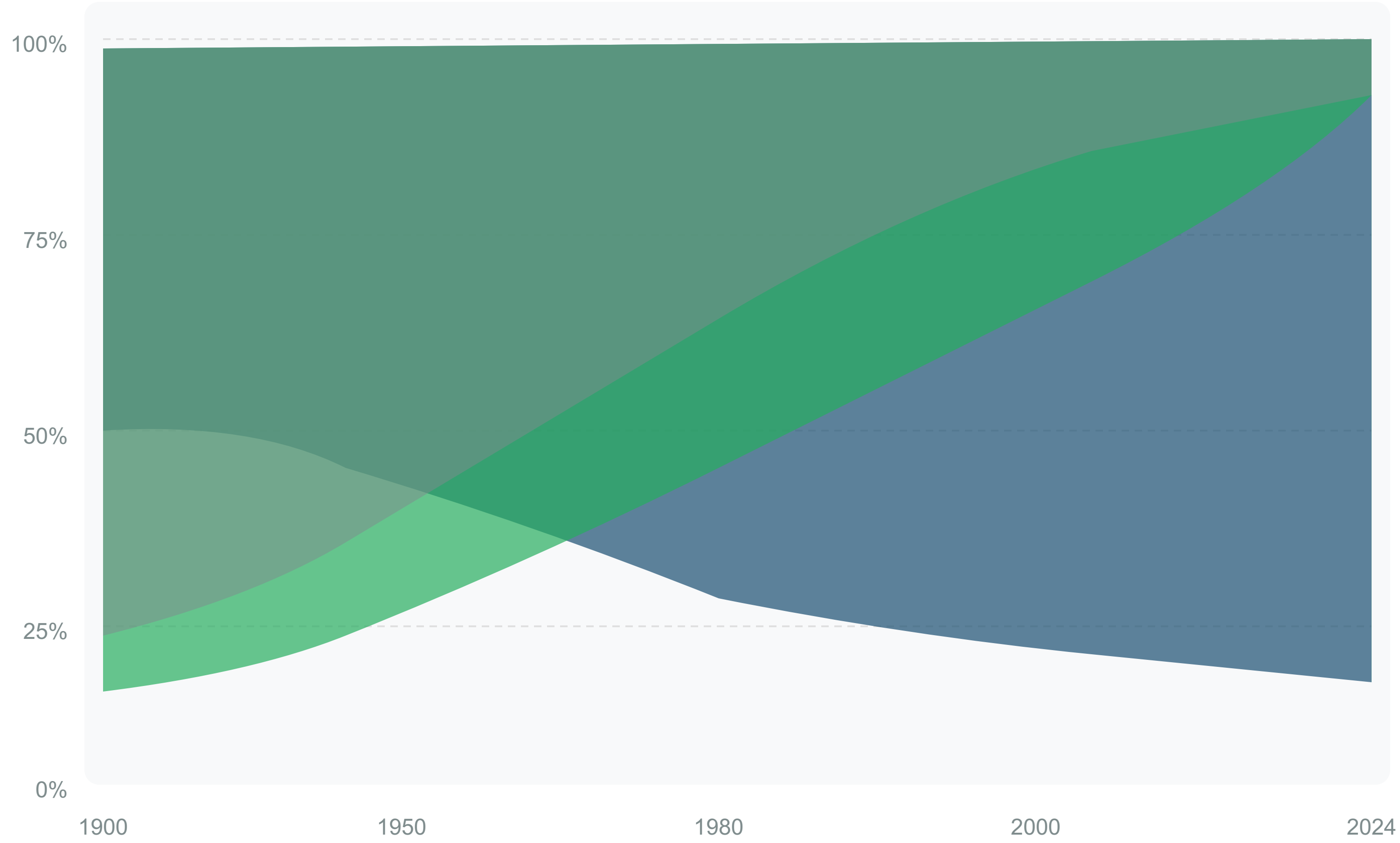
Growth is Slowing

But emissions have NOT yet peaked
Global action remains insufficient

Source: Global Carbon Budget 2025, Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Regional shares of global CO_2 emissions over time



Regional Legend

- Europe + United States
- Asia (led by China)
- Rest of World

Key Statistics

1900 Share:
>90% Europe + US

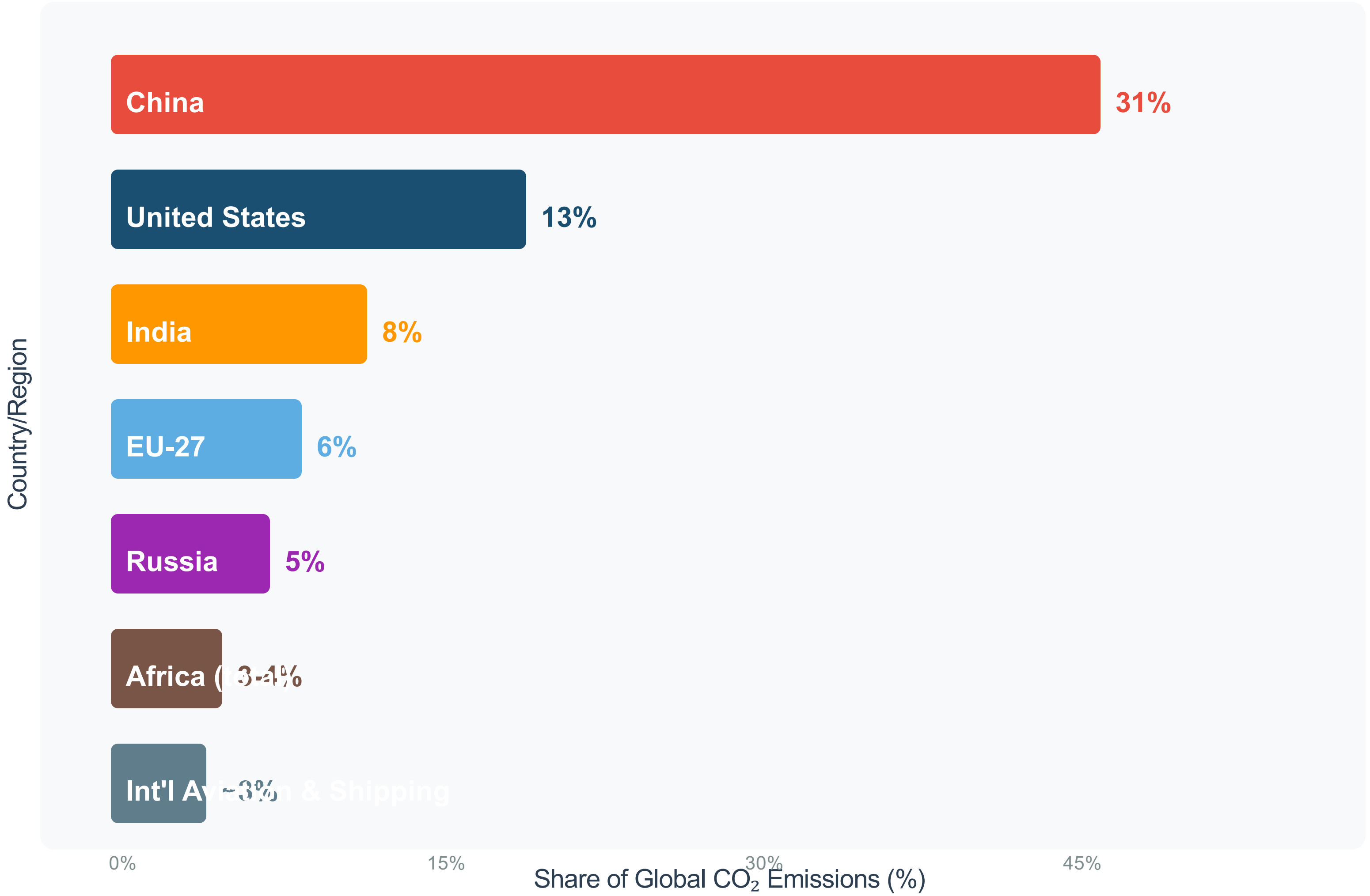
2024 Share:
<33% Europe + US

Asia now produces ~50% of global emissions

Source: Global Carbon Budget 2025, Our World in Data

China Now Emits More Than One-Quarter of Global CO_2

Country-level shares of global CO_2 emissions, 2024



Key Insight

China emits more than the next 4 largest emitters COMBINED

(US + India + EU-27 + Russia)

Historical Context

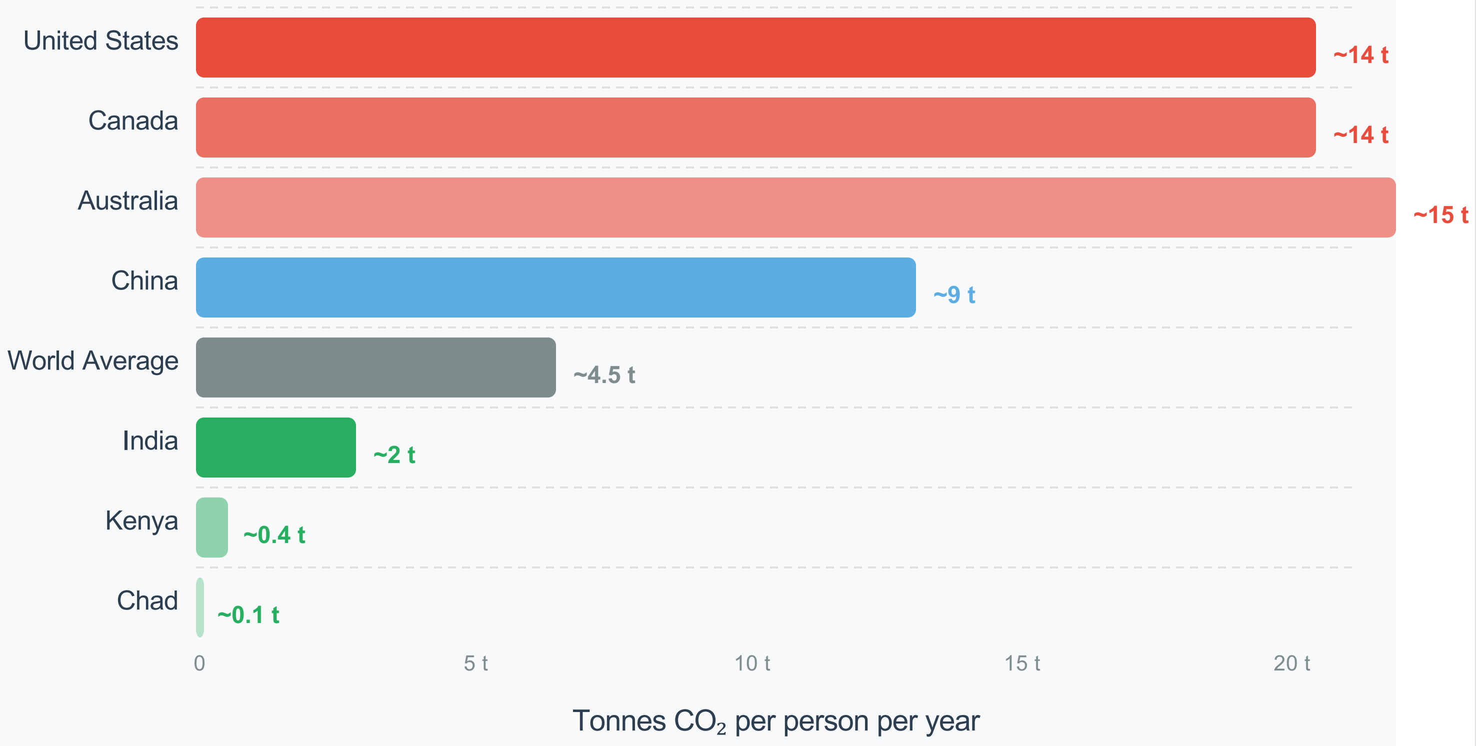
- China was a marginal emitter before 1980
- Rapid industrialization since 2000 drove dramatic rise
- Growth rate now slowing but absolute emissions still rising

Note: Africa and South America each account for only 3-4% — comparable to international aviation & shipping

Source: Global Carbon Budget 2025, Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

A 150x gap between lowest and highest emitting nations



The 150x Gap

USA/Canada
~14 t
per person/year

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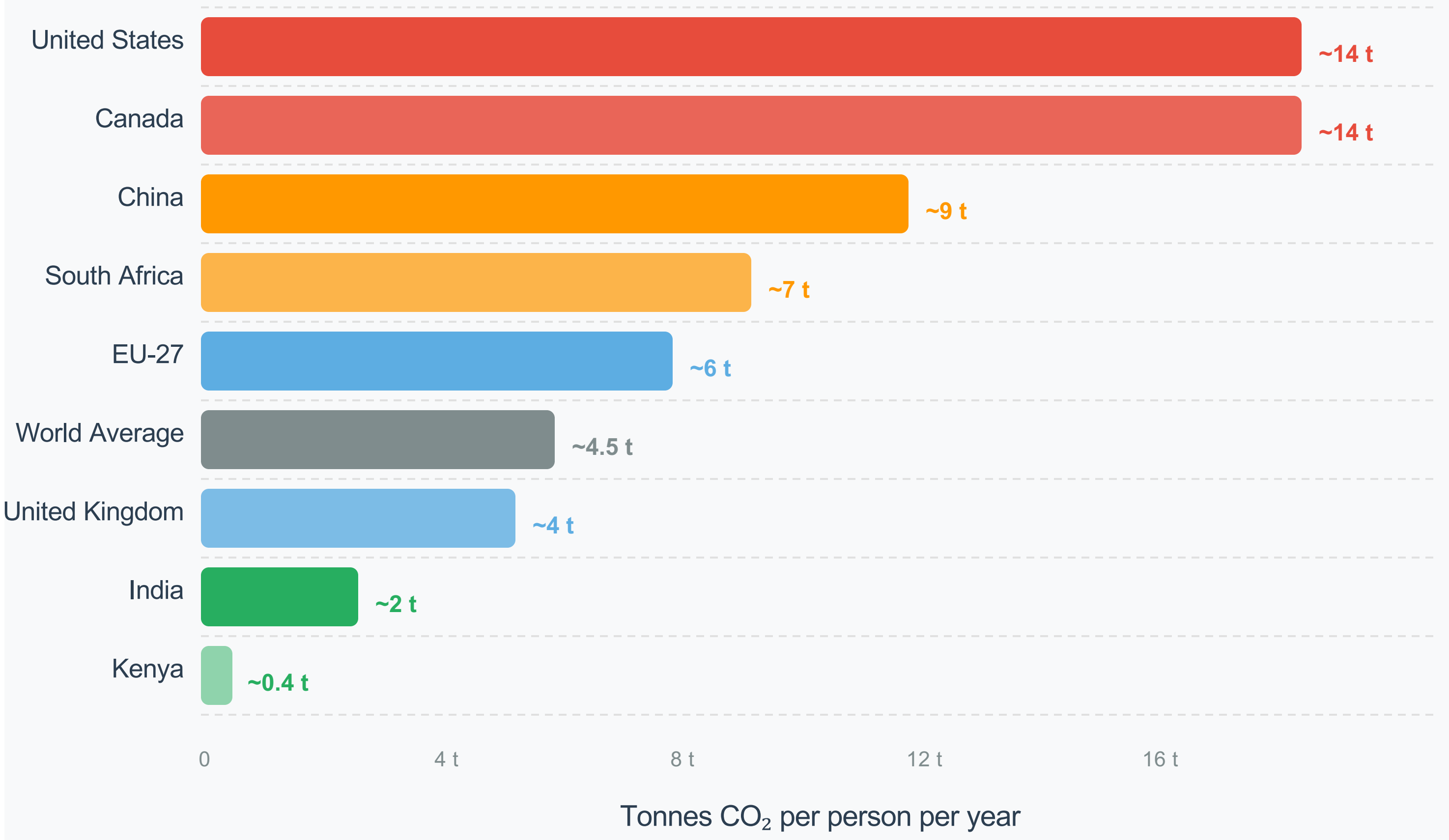
Chad/Niger
~0.1 t
per person/year

"The average American emits in **under 2 days**what the average person in Mali or Niger emits in an**entire year.**"

Note: Highest per capita emitters are oil-producing nations (Qatar, UAE, Bahrain, Kuwait) with small populations, not shown in chart above.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Current per capita CO_2 emissions for key countries (2024)



Key Observations

China: Largest Emitter, Below US Per Capita

Despite emitting 31% of global CO_2,

India: ~2 t per capita

3.5 billion people at this level would still produce far less than US alone

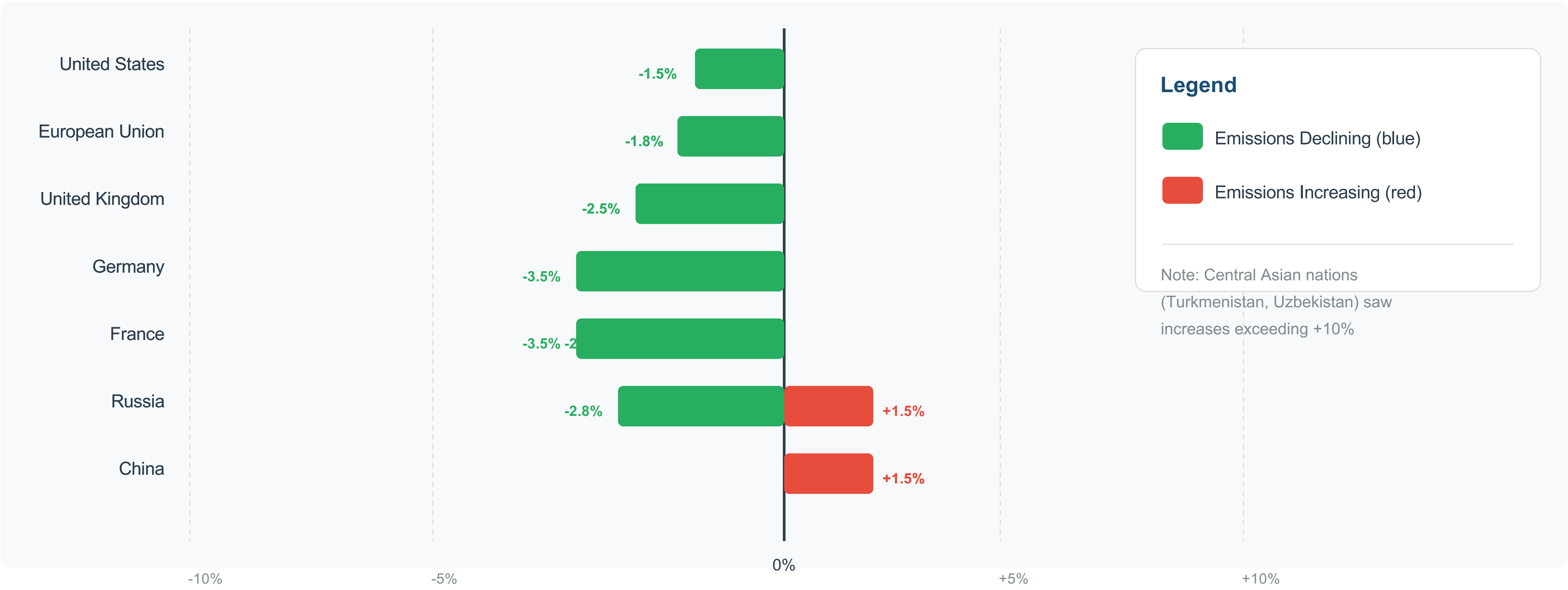
Color Legend

- High (>10 t) - US, Canada
- Medium (5-10 t) - China, S. Africa
- Moderate (3-6 t) - EU-27, UK
- Low (<3 t) - India, Kenya
- Global Average ~4.5 t

Source: Global Carbon Budget 2025, Our World in Data

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

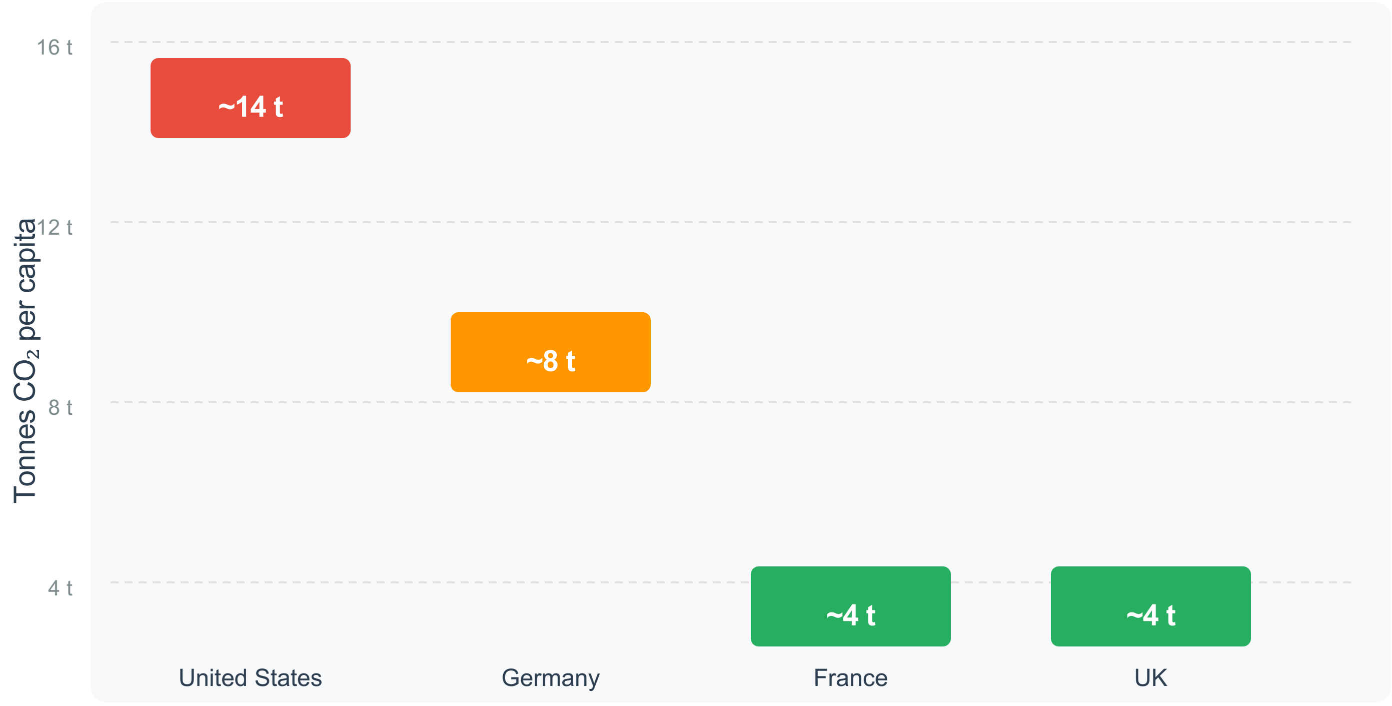
Annual percentage change in CO_2 emissions by region/country



Source: Global Carbon Budget 2025, Our World in Data

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the

Among wealthy nations with similar prosperity levels, per capita emissions vary significantly



Per Capita CO_2 Emissions (2024)

Why the Difference?

Energy Mix is the Key Factor

Countries with similar prosperity levels can have very different per capita emissions based on their energy mix choices.

Nuclear & Renewable-Rich Nations:

- France: ~70% nuclear electricity
- Portugal: High renewable share

→ Result: Low per capita emissions (~4 t)

Fossil Fuel-Dependent Nations:

- Germany: Heavy coal + natural gas
- Netherlands: High gas reliance
- United States: Coal + oil + gas

Key Insight: Policy and technology choices can decouple prosperity from emissions

Source: Global Carbon Budget 2025, Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

1

No Peak Yet

Global CO_2 emissions from fossil fuels exceed 35 billion tonnes per year and have not yet peaked.

2

China Leads but Per Capita Remains Below US

China is the largest annual emitter (~31%) but per capita emissions (~9 t) remain below the US (~14 t).

3

Historical Responsibility Dominated by Europe & US

Europe and the US accounted for >90% of historical emissions; now less than one-third combined.

4

150x Per Capita Gap Persists

A 150x gap exists between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and highest (US/Canada, ~14-15 t/person).

5

Energy Policy Can Decouple Prosperity from Emissions

Countries like France and the UK show that similar prosperity can coexist with much lower emissions through nuclear and renewable energy.

Source: Global Carbon Budget 2025, Our World in Data